

Altermagnetic Flatband-Driven Fermi Surface Geometry for Giant Tunneling Magnetoresistance

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这篇论文讲述了一个关于**如何让电脑内存变得更快、更省电**的突破性发现。为了让你轻松理解，我们可以把这篇复杂的科学文章想象成在寻找一种**“超级高速公路”**的设计方案。

1. 背景：我们需要什么样的“高速公路”？

现在的电脑内存（MRAM）就像是一个巨大的停车场，数据就是停在那里的车。

- **传统方案（铁磁体）**：就像用巨大的磁铁把车吸住。但这有个问题，磁铁会互相干扰（杂散磁场），导致停车场越来越拥挤，而且容易出错。
- **旧方案（反铁磁体）**：就像让车两两配对，头对头停，互相抵消磁力。这样很安静，不会互相干扰。但问题是，这种配对太完美了，导致很难分辨哪辆车是哪辆（自旋极化低），就像在黑暗中很难看清车牌，读写数据很慢。

新主角：交替磁性（Altermagnetism）

这篇论文介绍了一种“完美中间人”——**交替磁性材料**。

- 它像反铁磁体一样，两两配对，**没有杂散磁场**（不互相干扰）。
- 但它又像铁磁体一样，**能清晰地区分方向**（自旋分裂），让数据读写变得飞快。

2. 核心难题：如何把“路”堵死？

在隧道结（MTJ）这种设备里，我们要利用“隧道效应”来传输数据。

- **平行状态（P）**：两个电极的方向一致，车可以顺畅通过（电阻小，代表“1”）。
- **反平行状态（AP）**：两个电极方向相反，我们希望**车完全过不去**（电阻极大，代表“0”）。

问题出在哪里？

在大多数材料里，即使方向相反，总有一些“小路”或者“缝隙”能让车溜过去（这叫费米面重叠）。只要有一辆车溜过去，电阻就不会无限大，“0”和“1”的区别就不够明显，数据就容易出错。

3. 论文的发现：寻找“平坦的绝壁”

作者们研究了三种新材料（ V_2Te_2O 、 RbV_2Te_2O 、 KV_2Se_2O ），发现了一种神奇的几何结构，他们称之为**“平带”（Flatband）**。

用比喻来解释：

想象费米面（电子能跑的路）是一个地形图。

- **普通材料**：地形是起伏的山丘。即使方向相反，山丘之间总有低洼的谷地相连，车总能找到路溜过去。
- **普通交替磁性**：地形变成了两个分开的岛屿，但岛屿边缘还是有点接触（像圆弧一样），车还能勉强游过去。
- **这篇论文发现的 KV_2Se_2O** ：地形变成了**两个高耸的悬崖**，中间只有**四个极小的点**（像针尖一样）是连着的。这就叫**“平带驱动的费米面几何”**。因为路被限制在极小的“针尖”上，绝大多数车在反平行状态下**根本找不到路**，直接被“堵死”了。

4. 惊人的效果：从“大”到“巨大”

由于这种“悬崖式”的设计，KV₂Se₂O 材料表现惊人：

- **裸奔状态（真空层）**：即使不添加任何阻挡，它的“堵路”效果就已经让电阻差异达到了 **4300%**（普通材料通常只有几百）。
- **穿上“防弹衣”（加上绝缘层）**：作者们给它加了一层像 PbO（氧化铅）这样的绝缘墙。这就像在悬崖边又加了一道门。结果：电阻差异直接飙升至 **1,100,000%**（106%）！**对比**：目前最好的传统电脑内存（Fe/MgO/Fe）大约是 3700%。这个新材料比它们强了几百倍。

5. 方向很重要：选对“车道”

论文还发现了一个有趣的细节：这种材料很“挑食”。

- 如果你让车沿着 **[001]** 方向跑（垂直于层），就像在悬崖边跑，路被堵得死死的，TMR 值超高（100 万%）。
- 如果你让车沿着 **[100]** 方向跑（平行于层），就像在平地上跑，路就变宽了，TMR 值会下降（虽然还是比传统的好，但没那么夸张）。

这告诉我们，**设计这种设备时，必须像切豆腐一样，精确地切对角度，才能发挥最大威力。**

总结：这意味着什么？

这篇论文不仅仅是在实验室里发现了一种新材料，它提供了一套**全新的设计蓝图**：

1. **不再盲目寻找材料**：以前大家只是碰运气找磁性好的材料。
2. **开始“画地图”**：现在我们知道，只要找到那些具有***平坦能带**和**“针尖状费米面”***的层状材料，就能制造出超级高效的存储器。

一句话总结：

科学家发现了一种像“悬崖”一样的特殊材料，它能像一堵完美的墙，把不需要的电子流彻底堵死，从而制造出比现有电脑内存快几百倍、更省电的下一代存储设备。这就像是把原本泥泞的小路，改造成了只有单行道且设有自动闸机的超级高速公路。

这是一份关于论文《Altermagnetic Flatband-Driven Fermi Surface Geometry for Giant Tunneling Magnetoresistance》（交替磁性平带驱动的费用面几何结构实现巨隧穿磁电阻）的详细技术总结。

1. 研究背景与问题 (Problem)

- **背景**：磁阻随机存取存储器（MRAM）是下一代非易失性存储器的有力候选者。传统的铁磁（FM）隧道结（MTJ）存在杂散磁场，限制了器件密度和稳定性；而反铁磁（AFM）材料虽然无杂散场且动力学极快，但其自旋结构补偿导致能带中自旋极化几乎为零，难以用于自旋电子学。
- **新机遇**：“交替磁性”（Altermagnetism）作为一种新兴物态，结合了铁磁的自旋分裂能带结构和反铁磁的零净磁化特性，被视为 MRAM 的理想平台。
- **核心挑战**：在交替磁性隧道结（AMTJ）中，隧穿磁电阻（TMR）的大小取决于平行（P）和反平行（AP）配置下自旋极化传导通道的重叠程度。理想情况下，AP 配置下的自旋通道应完全不重叠以实现无限大的 TMR。然而，在大多数体块交替磁性材料中，由于费米面几何结构的限制，往往存在显著的自旋通道重叠（即“泄漏”电流），导致 TMR 难以达到理论极限。
- **研究目标**：探索如何通过费米面几何结构的调控（特别是利用平带效应），最小化自旋通道重叠，从而在实验已合成的交替磁性材料中实现超高性能的 AMTJ。

2. 研究方法 (Methodology)

- 材料选择:** 研究聚焦于三种实验已合成的体块交替磁性材料: V_2Te_2O 、 RbV_2Te_2O 和 KV_2Se_2O 。这些材料具有层状结构和高度各向异性的费米面。
- 理论计算: 电子结构计算:** 使用基于密度泛函理论 (DFT) 的 VASP 软件进行几何优化和能带结构计算。考虑了自旋轨道耦合和 Hubbard U 修正 (针对 V 和 Ti 轨道)。**量子输运模拟:** 结合非平衡格林函数 (NEGF) 方法, 使用 QuantumATK 软件模拟双探针器件系统的自旋相关输运性质。**模型构建:** 构建了 AMTJ 模型, 包括电极 (上述三种材料) 和绝缘势垒 (如真空、 PbO 、 TiO_2)。**参数分析:** 计算了不同晶向 ([001] 和 [100]) 下的自旋分辨透射系数、总透射谱以及 TMR 比率 (定义为 $TMR = (T_P - T_{AP})/T_{AP}$)。

3. 关键贡献与发现 (Key Contributions & Results)

A. 费米面几何结构与平带效应

- 平带驱动:** 研究发现, RbV_2Te_2O 和 KV_2Se_2O 在费米能级附近存在显著的**交替磁性平带** (Altermagnetic Flatbands)。这些平带导致费米面呈现准二维片状结构, 具有极强的各向异性。
- 自旋通道分离:** **V_2Te_2O :** 费米面在 Γ 点附近存在扩展的自旋筒并连续区, 导致 AP 状态下仍有大量泄漏电流, TMR 较低 (约 18%)。 **RbV_2Te_2O :** 费米面在动量空间中呈现弧状 (Arc-like) 的自旋筒并区域, 部分抑制了泄漏, TMR 提升至 435%。 **KV_2Se_2O (核心发现):** 得益于平带效应, 其费米面在 AP 配置下几乎完全分离, 仅在动量空间中留下**四个离散的节点状 (Nodal-like) 自旋筒并点**。这种几何结构极大地压缩了自旋筒并的相空间。

B. 巨隧穿磁电阻 (Giant TMR) 的实现

- 本征 TMR:** 在仅使用真空势垒的理想模型中, 基于 KV_2Se_2O 的 AMTJ 在费米能级处实现了高达 $4.3 \times 10^3\%$ 的本征 TMR。
- 引入绝缘势垒后的增强:** 为了模拟真实器件, 引入了与 KV_2Se_2O 晶格匹配良好的绝缘势垒 **PbO** (沿 [001] 方向)。由于 PbO 的能带结构与电极的动量选择性匹配, 进一步增强了平行态的透射, 同时保持了 AP 态的极低透射。最终实现了惊人的 $1.1 \times 10^6\%$ (即 1100 万%) 的 TMR 比率。这一数值比传统 $Fe|MgO|Fe$ 结的理论极限 ($\sim 3700\%$) 高出三个数量级, 也远超此前报道的 RuO_2 基 AMTJ。
- 晶向依赖性:** 研究还发现, 当电流沿 [100] 方向传输时, 由于对称性约束解除, 自旋筒并连续区重新出现, TMR 降至 $3.3 \times 10^3\%$ 。这证明了通过晶体取向工程可以调控器件性能。

C. 性能基准对比

论文将 KV_2Se_2O 基 AMTJ 的性能与现有的实验和理论 MTJ 结果进行了基准测试。结果显示, KV_2Se_2O 在保持室温奈尔温度 ($T_N > 300K$) 的同时, 提供了目前理论预测中最高的 TMR 值, 且无需依赖极低温或磁场诱导相变。

4. 研究意义 (Significance)

- 材料筛选的新范式:** 确立了**“平带驱动的费米面几何结构”**作为设计高性能交替磁性自旋电子器件的关键原则。即寻找具有准层状结构、能产生平带并导致费米面在动量空间中高度分离 (仅保留离散节点) 的材料。
- 突破 TMR 极限:** 证明了 KV_2Se_2O 是构建下一代超高 TMR 存储器件的极具竞争力的候选材料, 其性能指标远超现有铁磁和传统反铁磁方案。
- 器件设计指导:** 提出了通过晶体学取向控制 (如选择 [001] 方向) 和界面工程 (选择合适的势垒材料如 PbO) 来优化器件性能的具体策略。
- 理论机制阐明:** 深入揭示了费米面几何形状 (从连续区到弧状再到离散节点) 对自旋通道重叠及 TMR 性能的定量影响, 为理解交替磁性输运机制提供了清晰的物理图像。

总结：该工作通过第一性原理计算，发现并验证了 KV₂Se₂O 作为一种具有平带特征的交替磁性材料，能够通过其独特的费米面几何结构（节点状自旋简并）实现前所未有的巨隧穿磁电阻效应，为下一代高密度、低功耗、无杂散场的自旋电子存储器提供了重要的材料基础和理论蓝图。

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类似论文

Unraveling the Atomic-Scale Pathways Driving Pressure-Induced Phase Transitions in Silicon

本研究通过结合 GAP 势函数分子动力学、固态爬升弹性带（SS-NEB）计算及第一性原理验证，揭示了硅在压力诱导下从 BC8/R8 相向高密度相（hd）转变的原子尺度机制，阐明了纯退火条件下的局部形核过程及压力对形核势垒的影响，从而将实验观测与模拟结果建立了联系。

Fabrizio Rovaris, Anna Marzegalli, Francesco Montalenti + 1 more2026-03-06  cond-mat.mtrl-sci

Intrinsic higher-order topological states in 2D honeycomb Z₂ quantum spin Hall insulators

该研究通过第一性原理计算和紧束缚模型，发现二维蜂窝状 Bi、HgTe 及其在 Al₂O₃(0001) 衬底上的结构均展现出第一阶与高阶拓扑绝缘态的共存特性，其中 HgTe/Al₂O₃(0001) 因其实验可行性和低能角态而成为极具应用前景的材料。

Sibin Lü, Jun Hu2026-03-06  cond-mat.mtrl-sci

Sliding multiferrocity in van der Waals layered CrI₂

该研究通过第一性原理计算揭示了层状 CrI₂ 中由层间滑移驱动的铁电性与自旋驱动极化之间的强磁电耦合机制，并预测了其单层中可通过电场翻转自旋手性的二维多铁特性。

Hui-Shi Yu, Xiao-Sheng Ni, Kun Cao2026-03-06  cond-mat.mtrl-sci

Computing finite-temperature elastic constants with noise cancellation

本文通过推广一种原本用于压电耦合系数的噪声消除方法，在应用相同恒温方案模拟应变与参考系统后，成功实现了在热有序及无序体系中高精度计算有限温度下的弹性常数，有效克服了传统方法中热噪声大和强非谐效应带来的困难。

Debashish Mukherji, Marcus Müller, Martin H. Müser2026-03-06  cond-mat.mtrl-sci

Structure and magnetism of MnGe thin films grown with a nonmagnetic CrSi template

该研究利用非磁性 CrSi 模板在 Si(111) 衬底上成功生长了高质量 B20 结构 MnGe 薄膜，并发现其在低温下表现出与三重 Q 拓扑自旋 hedgehog 晶格或多畴单 Q 螺旋态相关的异常剩磁现象。

B. D. MacNeil, J. S. R. McCoombs, D. Kalliecharan + 7 more2026-03-06  cond-mat.mtrl-sci

Altermagnetic Flatband-Driven Fermi Surface Geometry for Giant Tunneling Magnetoresistance

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Altermagnetism, characterized by zero net magnetization and symmetry-protected spin-split band structures, has recently emerged as a promising platform for spintronics. In altermagnetic tunnel junctions (AMTJs), the suppression of tunneling in the antiparallel configuration relies on the mismatch between spin-polarized conduction channels in momentum space. However, ideal nonoverlapping spin-polarized Fermi surfaces are rarely found in bulk altermagnets. Motivated by the critical influence of Fermi surface geometry on tunneling magnetoresistance (TMR), we investigate three experimentally synthesized altermagnets – bulk V_2Te_2O , RbV_2Te_2O , and KV_2Se_2O – to elucidate how flatband-driven Fermi surfaces minimize spin-channel overlap and boost AMTJ performance. Notably, RbV_2Te_2O and KV_2Se_2O host flat altermagnetic Fermi sheets, which confine spin degeneracy to minimal arc-like or nodal-like regions. Such Fermi surface geometry drastically reduces spin overlap, resulting in an unprecedented intrinsic TMR well over $10^3\%$ in the KV_2Se_2O -based AMTJ. Incorporating an insulating barrier further enhances the TMR to $\sim 10^6\%$, surpassing most conventional MTJs. These results not only establish KV_2Se_2O as a compelling candidate AMTJ material, but also highlight the critical role of flatband Fermi surface geometry in achieving high-performance altermagnetic-spintronic device technology.

I. INTRODUCTION

Magnetoresistive random-access memory (MRAM) is a promising candidate for next-generation nonvolatile memory technology, owing to its fast operation speed and low power consumption [1–4]. A magnetic tunnel junction (MTJ) forms the core of MRAM, which comprises two ferromagnetic (FM) electrodes separated by an insulating barrier [5–7]. In conventional MTJs, binary information is encoded in the relative magnetization alignment of the FM electrodes, while data writing can be performed electrically through spin-transfer or spin-orbit torques and data reading is typically accomplished through the tunneling magnetoresistance (TMR) effect [8–10]. However, FM-based architectures suffer from stray magnetic fields, which limit device density and stability. Antiferromagnetic (AFM) materials inherently eliminate stray fields and exhibit ultrafast spin dynamics in the terahertz regime [11–13], but their compensated spin structure leads to negligible spin polarization in their band structures, which restricts their potential in spintronics.

Bridging the long-standing dichotomy between ferromagnetism and antiferromagnetism, altermagnetism has emerged as a compelling candidate for MRAM due to its “best-of-both-worlds” character: it features collinear AFM spin ordering together with FM-like spin-split electronic structures [14–17]. This unique combination enables spin-dependent transport in the absence of net magnetization, offering new opportunities for low-dissipation and high-speed spintronic applications. Unlike conventional antiferromagnets, altermagnets break both $\mathcal{PT}$ (space inversion and time reversal) and $\mathcal{T}t$ (time reversal combined with half-cell translation) symmetries, leading to sizable nonrelativistic spin splitting away from high-symmetry paths in the Brillouin zone (BZ) [18, 19]. These features make altermagnets a compelling building block for MTJs, combining the spin-polarized transport of ferromagnets with the stray-field-free stability characteristic of antiferromagnets [11, 20–23].

Although multiple altermagnets have been theoretically proposed, their experimental realizations remain limited. Among the reported candidates, RuO_2 remains one of the most extensively studied systems, but its altermagnetic nature is still under debate [24–28]. More recently, KV_2Se_2O [29–31] and RbV_2Te_2O [32] have been experimentally confirmed to host d -wave altermagnetic band structures through spin- and angle-resolved photo-

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toemission spectroscopy (SARPES). Their above-room-temperature Néel temperatures and highly anisotropic spin-polarized Fermi surfaces [33] render them particularly attractive for altermagnet-based spintronic applications.

The transport characteristics and overall performance of an altermagnetic tunnel junction (AMTJ) are critically influenced by the Fermi surface geometry of its electrodes [33, 34]. The extent of spin matching (or mismatching) between spin-polarized conduction channels directly determines the current magnitude in the parallel (P) or antiparallel (AP) state, and hence the tunneling magnetoresistance (TMR) – a key figure of merit for MRAM technologies. Motivated by the pivotal role of Fermi surface geometry in shaping TMR, we examine three experimentally realized altermagnets – bulk V_2Te_2O , RbV_2Te_2O , and KV_2Se_2O – to elucidate how altermagnetic flatbands can suppress spin channel overlap.

Our first-principles quantum transport simulations show that the quasi-layered altermagnets RbV_2Te_2O and KV_2Se_2O host altermagnetic flatbands near the Fermi level, which give rise to strongly anisotropic and quasi-2D Fermi surface sheets. As a direct consequence, intersections between opposite-spin channels are confined to arc-like or nodal-like regions in momentum space, which dramatically reduces the available phase space for spin degeneracy. This flatband-induced suppression of spin channel overlap strongly diminishes the AP-state tunneling, enabling the realization of large TMR ratios. While minimizing spin-channel overlap is a well-established paradigm for achieving large TMR, this work identifies altermagnetic flatbands as a concrete, material-specific mechanism for realizing this condition, thereby providing a viable and materials-guided route to high-performance AMTJs.

In particular, KV_2Se_2O exhibits only four spin-degenerate, nodal-like overlaps at the Fermi level, yielding an intrinsic TMR as high as $4.3 \times 10^3\%$ with a vacuum barrier. With an appropriate insulating barrier, the TMR can be further increased to $1.1 \times 10^6\%$, exceeding the theoretical value of conventional $Fe|MgO|Fe$ MTJs ($\sim 3700\%$) by three orders of magnitude [35, 36]. In contrast, when transport is oriented along a crystallographic direction with larger opposite-spin overlap, the TMR of the KV_2Se_2O AMTJ drops substantially to $3.3 \times 10^3\%$. These findings underscore the central role of Fermi surface engineering, particularly through altermagnetic flatband design and crystallographic orientation control, in achieving high-TMR AMTJs. This work not only reveals KV_2Se_2O as a compelling candidate for AMTJs, but also provides a blueprint for high-performance altermagnetic spintronic architectures, uniquely enabled by the spin-contrasting flatband Fermi surface geometry of quasi-layered altermagnets.

II. RESULTS

A. Fermi Surface Geometry in Magnetic Tunnel Junctions

The transport properties of MTJs are governed by spin-dependent quantum tunneling across a thin insulating barrier. The efficiency of this process is strongly dictated by the Fermi surface geometry of the FM electrodes, which determines the momentum-resolved conduction channels available for tunneling (Figure 1) [37, 38]. In the ballistic regime, electron transmission across the barrier requires the conservation of both spin and transverse momentum [39]. Because the Fermi surfaces of the FM electrodes are spin-polarized, the number of available conduction channels differs for majority and minority spins. As a result, the junction resistance depends sensitively on the relative magnetic alignment of the electrodes. When the magnetizations are parallel (P), the majority-spin channels on both sides align, yielding high conductance (low resistance). Conversely, in the antiparallel (AP) configuration, the strong mismatch between majority and minority spin bands significantly reduces the available tunneling pathways, leading to a high-resistance state. This resistance modulation via magnetic alignment leads to a TMR effect along all crystallographic directions in an FM-based MTJ [Figure 1(a)].

In the case of altermagnetic electrodes, strongly anisotropic, spin-polarized Fermi surfaces give rise to momentum-dependent and spin-contrasting conduction channels for opposite spins [19, 40]. When the Néel vectors of the two altermagnetic electrodes are antiparallel, the conduction channels become spin-mismatched, leading to a strong suppression of AP-state tunneling and thus producing a TMR effect [41]. For a typical planar d -wave Fermi surface geometry [Figure 1(b)], the rotational symmetry connecting opposite-spin sublattices enforces spin degeneracy at the Γ point, yielding a sizable number of spin-degenerate conduction channels. Consequently, AP-state transmission cannot be fully suppressed, and this residual “leakage” current limits the achievable TMR in such AMTJs.

Ideally, if the spin-resolved Fermi surfaces of the two electrodes exhibit no overlap at any transverse momentum [Figure 1(c)], transmission in AP state can be completely suppressed. C-paired spin–valley locking (SVL) in two-dimensional (2D) lattices provides a symmetry-driven route to realize such well-separated spin-polarized channels [42]. However, as interlayer coupling and the restoration of inversion symmetry in multilayer systems tend to hybridize opposite-valley spin states, SVL is typically realized in monolayer systems [43–46]. This makes direct implementation of SVL in AMTJs challenging, as stable vertical tunneling structures generally require bulk electrodes rather than monolayer components.

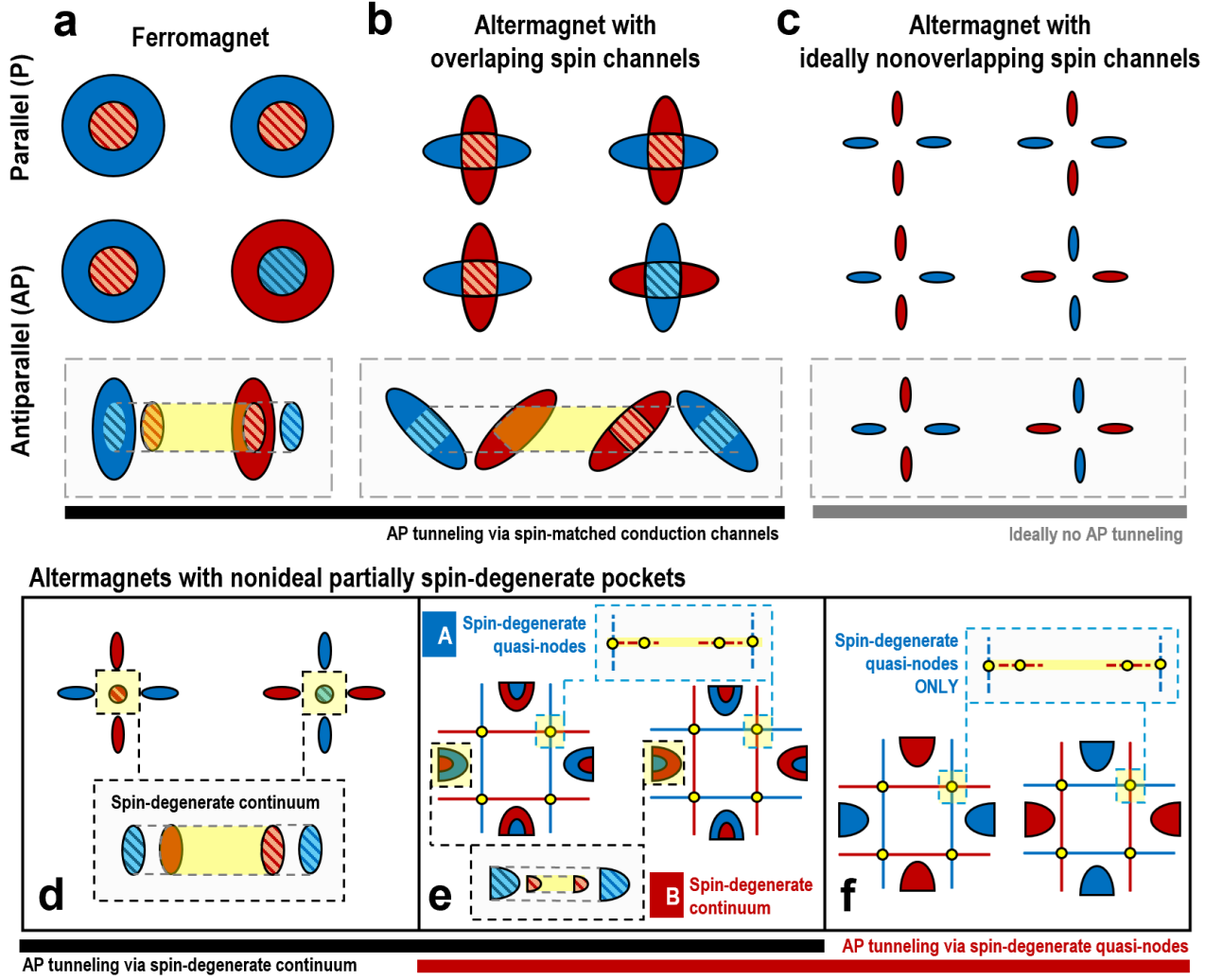


FIG. 1. Schematic illustration of tunneling magnetoresistance (TMR) mechanisms in MTJs with different electrode materials. a) Conventional ferromagnets. b) Typical altermagnets with overlapping spin conduction channels. c) AMs with ideally nonoverlapping spin conduction channels. (d-f) AMs with nonideal partially spin-degenerate Fermi pockets that allow residual AP tunneling. Two types of overlap are considered: spin-degenerate continuum and discrete spin-degenerate quasi-nodes. Red and blue denote spin-up and spin-down conduction channels, respectively.

B. Candidate Altermagnets for Magnetic tunnel Junctions

As fully separated spin-polarized pockets are rarely realized in three-dimensional (3D) bulk materials, we focus on more experimentally accessible cases of bulk AMs that host partially spin-degenerate conduction channels [47, 48]. Achieving large TMR in such systems requires Fermi surface geometries with minimal overlap between opposite-spin conduction channels. In particular, sufficiently flat Fermi surfaces can restrict this overlap to isolated, nodal-like spin-degenerate regions. Motivated by this consideration, we examine three experimentally synthesized layered altermagnetic candidates – bulk V_2Te_2O [49], RbV_2Te_2O [32] and KV_2Se_2O [29]

– each of which hosts highly anisotropic, band-flattened Fermi surfaces. As illustrated in Figures 1(d–f), these systems exhibit distinct forms of residual overlap, ranging from extended spin-degenerate continua to discrete spin-degenerate quasi-nodes. In the following, we systematically investigate their structural and electronic characteristics (see Figure 2) to assess their suitability for AMTJ device applications.

Monolayer V_2Te_2O exhibits a tetragonal lattice with a checkerboard configuration [Figure 2(a)]. Each vanadium atom is coordinated by four Te and two O atoms in a distorted octahedral environment, and exhibits staggered out-of-plane magnetization. In the multilayer form, V_2Te_2O adopts an AB-AFM stacking, where adjacent layers are laterally shifted by half a unit cell and alternate in spin orientations [49]. Structurally, the two

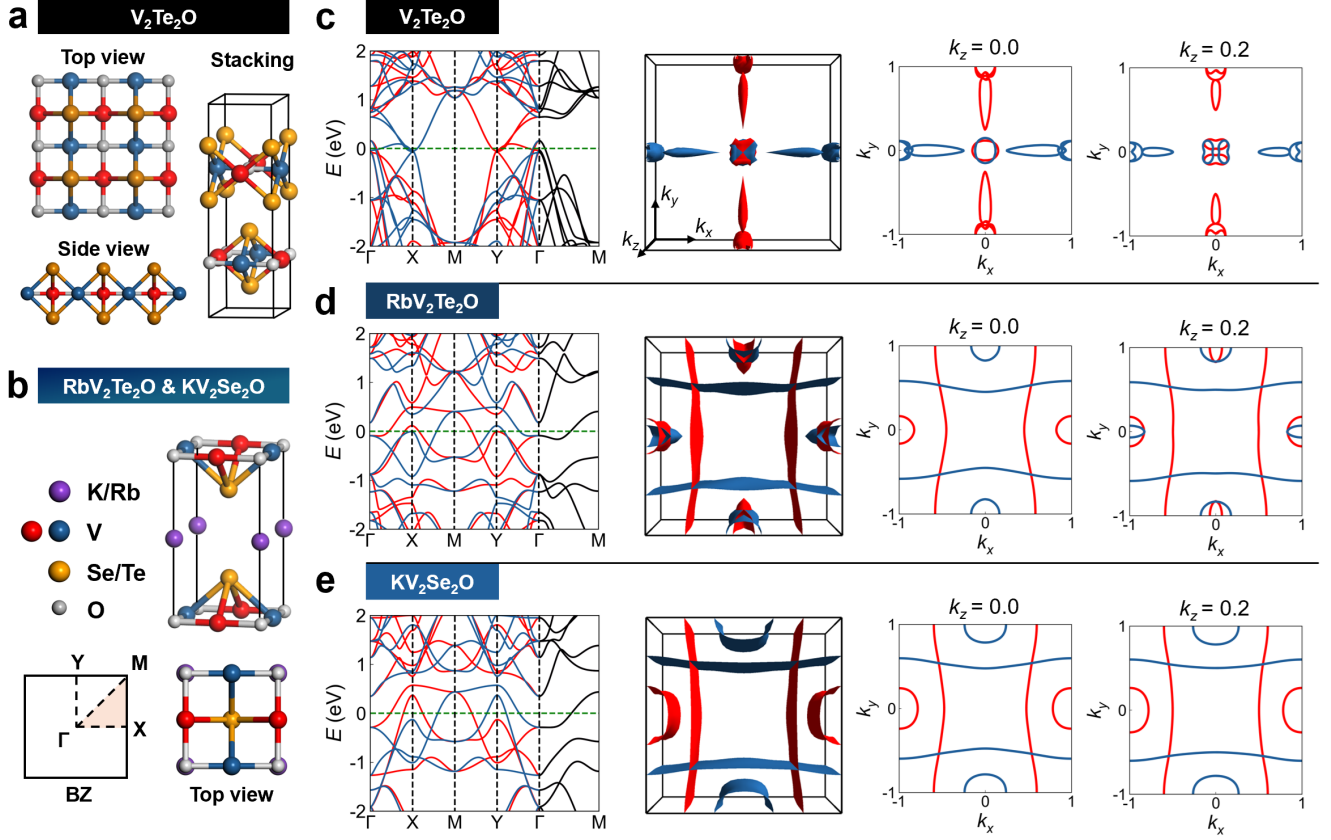


FIG. 2. Crystal and magnetic structures of a) V_2Te_2O , b) RbV_2Te_2O and KV_2Se_2O . Red and blue indicate up and down spins of the V atoms, respectively. (c-e) Electronic band structures, three-dimensional spin-resolved Fermi surfaces and Fermi surface cross-sections at different k_z planes ($k_z = 0$ and 0.2) of different materials. Spin-degenerate bands are shown in black.

opposite-spin sublattices are not related by spatial inversion or translation, but are instead connected by C_{4z} or $M_{1\bar{1}0}$ operation. This symmetry arrangement gives rise to a planar d -wave altermagnetic order, which is reflected in its spin-split band structures. The energy bands along the $\Gamma-X-M$ and $\Gamma-Y-M$ paths exhibit opposite spin splitting [Figure 2(c)], whereas those along the $\Gamma-M$ path remain spin-degenerate. This momentum-dependent spin texture results in discrete spin-polarized Fermi pockets near the X and Y valleys, whereas multiple band crossings near the Γ point generate an extended spin-degenerate continuum at the Fermi level.

RbV_2Te_2O and KV_2Se_2O are experimentally confirmed metallic d -wave altermagnets [29, 32]. These altermagnets share an identical layered structure, consisting of $V_2Te(Se)_2O$ sheets separated by Rb/K atoms [Figure 2(b)]. They crystallize in a tetragonal lattice with space group P_4/mmm . Nuclear magnetic resonance measurements reveal staggered spin orientations of V atoms along the c axis within each $V_2Te(Se)_2O$ plane, with a Néel temperature well above room temperature. The magnetic structure also breaks both $\mathcal{PT}$ and $\mathcal{Tt}$ symmetries, while preserving combined symmetries such as $[C_2||C_{4z}]$ and $[C_2||M_{1\bar{1}0}]$ (the left operation acting in spin space

while the right acting in real space). These symmetry characteristics produce spin-split patterns analogous to those of bulk V_2Te_2O [Figures 2(d,e)]. The spin-resolved Fermi surfaces of both compounds consist of quasi-2D sheets extending along k_z , with distinct pockets near the Brillouin zone (BZ) boundaries. A comparison of the $k_z = 0.2$ Fermi surface cross-sections reveals that RbV_2Te_2O retains arc-like regions of residual overlap, while KV_2Se_2O achieves nearly complete spin separation, with only four spin-degenerate nodal points. The presence of such minimal spin-degenerate regions suggests strong suppression of AP-state leakage current and, consequently, large TMR in these altermagnetic systems.

The emergence of the altermagnetic flatbands observed in these two materials can be traced to a pronounced directional anisotropy in the electronic dispersion [33] (See Section B of the Supporting information for details). As shown in Figure S3, the band giving rise to the quasi-2D Fermi surface sheets in both RbV_2Te_2O and KV_2Se_2O is dominated by the $V-d_{xz}$ orbital for spin-up electrons and by the $V-d_{yz}$ orbital for spin-down electrons. This orbital selectivity induces a strong spin-dependent crystallographic anisotropy between the x and y directions. Moreover, owing to the intercalation of Rb/K atoms,

electronic coupling along the z direction is intrinsically weaker than that within the xy plane. As a result, altermagnetic flatbands emerge in these layered compounds, leading to minimal spin channel overlap at the Fermi level.

C. Intrinsic transport properties of V_2Te_2O , RbV_2Te_2O and KV_2Se_2O AMTJs

Having established the Fermi surface geometries of the three representative bulk altermagnets, we now turn to their intrinsic transport properties along the $[001]$ direction. As a preliminary assessment, the AMTJs are modeled using an ideal 6 Å vacuum spacer [Figure 3(a)]. Such modelling setup enables a more direct assessment on how Fermi surface geometry inherent to each material influences the resulting TMR performance, before addressing the complicated interfacial effects when an insulating barrier is employed. Figures 3(b-d) show the transverse-momentum ($k_{\parallel}$)-resolved transmission coefficients at the Fermi level for spin-up and spin-down electrons. In the P state, the transmission profiles closely resemble the projections of the anisotropic Fermi surfaces onto the $[001]$ plane, as the spin-matched bands of the two electrodes generate efficient conduction channels. In contrast, electron transmission in the AP state is strongly suppressed throughout the BZ, except in several regions where spins are degenerate, namely (i) an extended continuum for V_2Te_2O [Figure 3(b)]; (ii) arc-like features for RbV_2Te_2O [Figure 3(c)]; and (iii) nodal-like features for KV_2Se_2O [Figure 3(d)].

The V_2Te_2O -based AMTJ exhibits pronounced transmission features for both spin channels in the P state, as characterized by a broad high-intensity region centered at the Γ point and extended continua along the $-X-\Gamma-X$ or $-Y-\Gamma-Y$ direction. The spin-up and spin-down transmission patterns share identical geometries, differing only by a $\pi/2$ rotation, which is consistent with the d -wave symmetry of the altermagnetic order. Consequently, the total current along the $[001]$ direction remains effectively spin-neutral. Owing to the large spin-degenerate continuum around the Γ point, AP-state tunneling cannot be completely suppressed, giving rise to a residual leakage current.

The evolution of the total transmission with energy further reveals the underlying mechanism [Figure 3(e)]. In both configurations, the transmission initially decreases with increasing energy. Beyond $E > 0.16$ eV, however, the total transmission in the P state T_P ($T_P = T_P^{\uparrow} + T_P^{\downarrow}$) exhibits a sharp increase while T_{AP} drops abruptly to nearly zero (shaded region). This sudden contrast originates from the evolution of the bulk V_2Te_2O Fermi surfaces at higher energies. As illustrated in Figure 4, increasing the energy from 0.06 to 0.18 eV progressively reduces the spin-degenerate continuum near the Γ point, which eventually disappears at $E = 0.18$ eV. At this energy, V_2Te_2O becomes an ideal

electrode material for AMTJs, exhibiting perfectly separated spin-polarized conduction channels. The energy-resolved TMR ratio shown in Figure 3(e) is defined as $TMR = (T_P - T_{AP})/T_{AP}$. Owing to the substantial spin channel overlap around the Γ point, the V_2Te_2O -based AMTJ yields a modest TMR of only 18% at E_F .

In contrast to V_2Te_2O , the Fermi surface geometries of RbV_2Te_2O and KV_2Se_2O exhibit no overlap near the Γ point [Figures 3(c,d)]. Both AMTJs feature narrow transmission channels in the P state, with distinct Fermi pockets at the BZ boundary, which is consistent with their quasi-2D altermagnetic nature. However, the spin-polarized pockets in RbV_2Te_2O are not completely separated, giving rise to several extended spin-degenerate “arcs” that sustain AP transmission. By contrast, the KV_2Se_2O -based AMTJ hosts only four discrete, “nodal-like” intersection points between opposite-spin conduction channels. These isolated nodes, providing minimal conduction pathway, immediately suggest a strong suppression of AP-state tunneling in the device.

The distinction between the spin-degenerate “arcs” in RbV_2Te_2O and the discrete “nodal points” in KV_2Se_2O is quantitatively reflected in their total transmission spectra and the corresponding TMR ratios [Figure 3(f,g)]. In RbV_2Te_2O , the persistence of spin-degenerate “arcs” leads to a moderate TMR ratio of 435% at the Fermi level. In contrast, the KV_2Se_2O -based AMTJ exhibits a pronounced difference between the P- and AP-state transmissions at E_F , resulting in a much higher TMR of $4.3 \times 10^3\%$. The shaded region in Figure 3(f) highlights a reduction in AP tunneling for RbV_2Te_2O , which can be attributed to subtle changes in its Fermi surface geometry (see Section D of the Supporting Information for details).

Despite the broader spin channel overlap in V_2Te_2O , the absolute values of the AP-state transmission coefficients at specific k -points are significantly larger in the RbV_2Te_2O device [Figure 3(c)]. For example, $T_{AP}^{\uparrow}$ near the Γ point reaches $\sim 10^{-7}$ in the V_2Te_2O AMTJ, whereas $T_{AP}^{\uparrow}$ at the arc-like features along the $\Gamma-X$ direction in the RbV_2Te_2O device is on the order of 10^{-5} . This pronounced disparity in the magnitude of the transmission coefficients ultimately leads to a larger total AP-state transmission at the Fermi level in the RbV_2Te_2O MTJ.

Taken together, the contrasting forms of the spin-degenerate regions – continua in V_2Te_2O , arcs in RbV_2Te_2O , and discrete nodal points in KV_2Se_2O – together with their corresponding order-of-magnitude variations in the TMR, underscore the pivotal role of Fermi surface overlap in determining and pushing the performance limits of AMTJs.

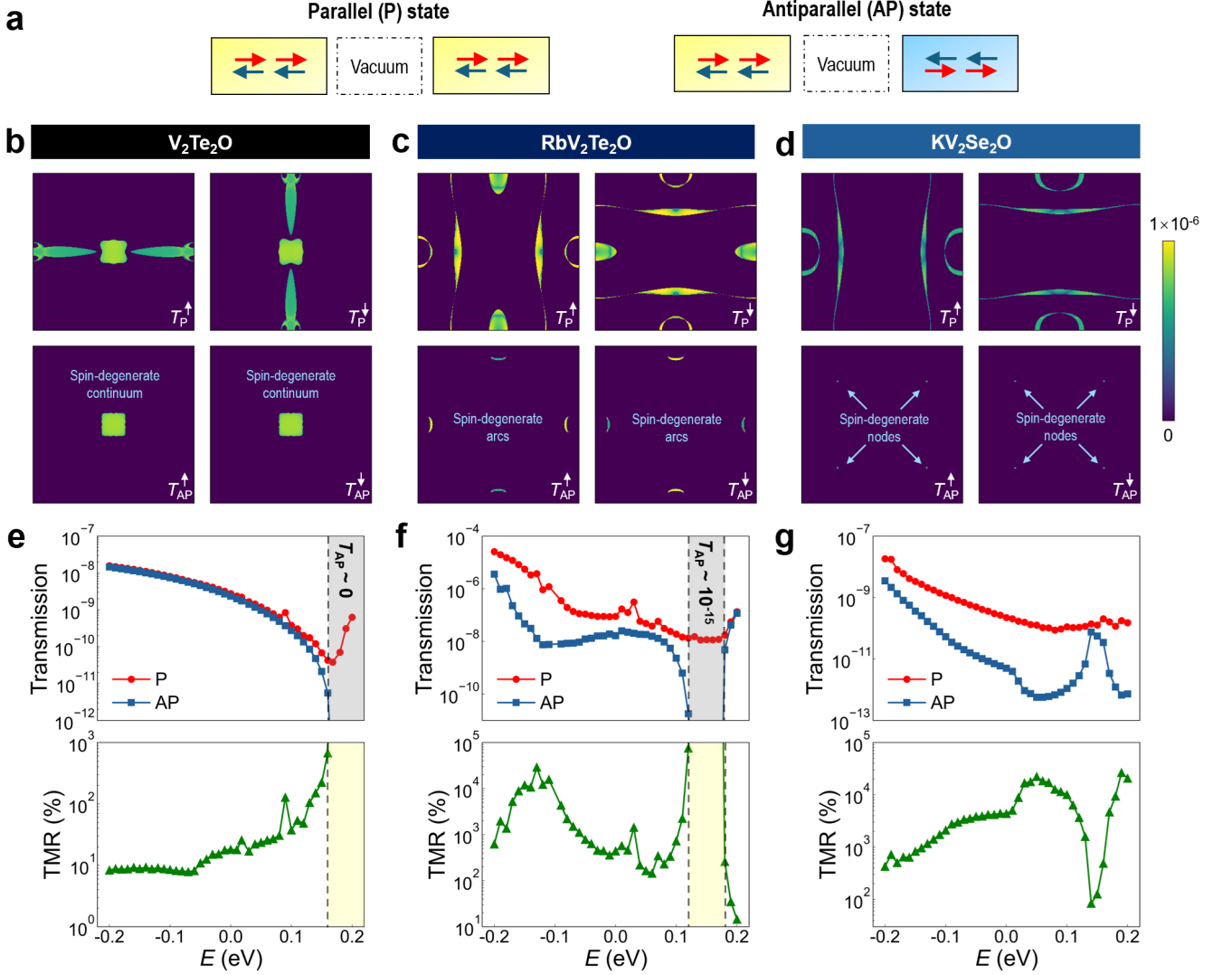


FIG. 3. Intrinsic transport properties of AMTJs with vacuum barriers based on V_2Te_2O , RbV_2Te_2O and KV_2Se_2O electrodes. a) Schematic illustration of magnetization alignments in the parallel (P) and antiparallel (AP) configurations. (b-d) Calculated spin-resolved, $k_{\parallel}$ -resolved transmission coefficients at the Fermi level for each AMTJ in both P and AP states. (e-g) Energy-dependent total transmission spectra for P and AP states, along with the resulting TMR ratio. Shaded regions highlight the energy ranges where AP tunneling T_{AP} is significantly suppressed due to changes in available conduction channels when energy is away from the Fermi level. The normalized TMR ratios, defined as $(T_P - T_{AP})/T_P$, are provided in Figure S4 of the Supporting Information.

D. Tunneling magnetoresistance of KV_2Se_2O -based AMTJ with insulating spacers

Having identified KV_2Se_2O as the most promising electrode material with intrinsically high TMR at the Fermi level, we next investigate its performance in more realistic device architectures that incorporate an insulating barrier. The choice of barrier can critically influence both tunneling efficiency and spin selectivity, as its electronic structure and symmetry compatibility with the electrodes directly affect the available tunneling pathways [50]. An appropriately chosen barrier can therefore substantially enhance the achievable TMR ratio beyond

the intrinsic limit imposed by the electrode Fermi surface. To this end, we construct KV_2Se_2O -based AMTJs incorporating realistic barrier materials along the [001] and [100] directions, and systematically evaluate their transport characteristics.

For transport along the [001] direction, we select PbO as the tunnel barrier owing to its excellent structural compatibility with KV_2Se_2O and a small lattice mismatch of only 1.1%. PbO is a well-established van der Waals semiconductor with a tetragonal crystal structure [51], as illustrated in Figure 5(a). A monolayer of PbO consists of an oxygen atomic layer sandwiched between two lead layers arranged in a checkerboard lattice. In

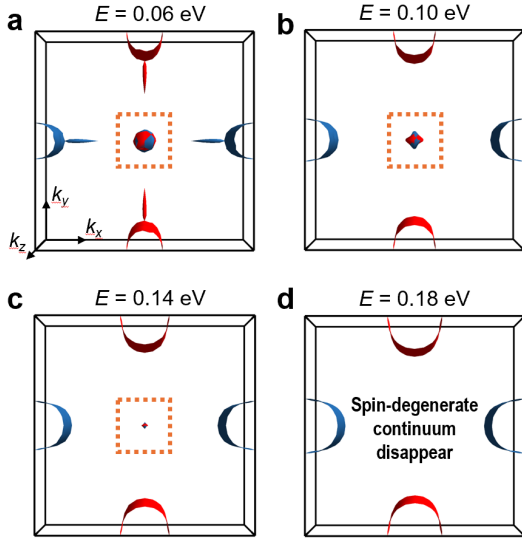


FIG. 4. Three-dimensional Fermi surfaces of bulk V_2Te_2O at a) $E = 0.06$ eV, b) $E = 0.10$ eV, c) $E = 0.14$ eV and d) $E = 0.18$ eV. Red and blue indicate spin-up and spin-down Fermi surfaces, respectively. The spin-degenerate continuum near the Γ point progressively diminishes with increasing energy and entirely vanishes at $E = 0.18$ eV.

multilayer form, it adopts an AA-stacking configuration, in which adjacent PbO layers are aligned vertically with an interlayer spacing of 4.97 Å. Owing to the absence of magnetic elements in its unit cell, bulk PbO exhibits spin-degenerate electronic bands and a band gap of 1.23 eV [Figure 5(b)]. In our AMTJ design, a three-layer PbO barrier is employed to separate the two electrodes. The layer-resolved density of states (LDOS) [Figure 5(c)] reveals a well-defined band gap at the Fermi level within the barrier region, confirming that the PbO barrier effectively suppresses direct electronic coupling between the electrodes and ensures transport dominated by quantum tunneling.

The $k_{||}$ -resolved transmission coefficients for the KV_2Se_2O -based AMTJ along the [001] direction are shown in Figure 5(d). The transmission patterns in the P state closely resemble those of the intrinsic (vacuum-barrier) case, while the nodal-like features remain well preserved in the AP configuration. The total transmission in the P state is enhanced by nearly three orders of magnitude compared with the vacuum-barrier junction [Figure 5(e)], whereas the increase in the AP-state transmission is much smaller. Specifically, T_P rises from 2.2×10^{-10} to 1.4×10^{-7} , while T_{AP} increases only from 5.0×10^{-12} to 1.2×10^{-11} at the Fermi level. Consequently, a giant TMR ratio of $1.1 \times 10^6\%$ is achieved at E_F , exceeding the previously reported value of $\sim 500\%$ in $RuO_2|TiO_2|RuO_2$ AMTJs [21] by more than three orders of magnitude.

The enhancement of TMR upon introducing an appropriate tunnel barrier can be understood within the framework of tunneling theory [52]. Electron transmis-

sion through an insulating barrier decays exponentially as $e^{-\kappa d}$, where d denotes the barrier thickness and κ is the decay rate determined by the complex band structure of the barrier material [53]. As shown in Figure S6, the lowest decay-rate distribution in the band gap of PbO develops extended low- κ regions at larger in-plane momenta $k_{||}$, which closely coincide with the momentum-resolved conduction channels of the KV_2Se_2O electrodes. In contrast to the vacuum-barrier case, where the decay rate is nearly independent of $k_{||}$, this momentum-selective matching substantially enhances the total transmission in the parallel configuration, thereby leading to a pronounced increase in the TMR ratio.

Building on the momentum-selective tunneling mechanism discussed above, the use of an altermagnetic semiconductor as the tunnel barrier provides an additional route to further enhance the TMR via spin-dependent filtering effects [54, 55]. Unlike nonmagnetic semiconductors, altermagnetic semiconductors possess symmetry-enforced spin-split band structures, giving rise to evanescent states with different decay rates for spin-up and spin-down electrons [56]. With appropriate matching between the barrier and the electrodes, this spin-dependent decay can provide an additional contribution to the TMR effect [57, 58]. As discussed in Section F of the Supporting Information, we employ multilayer Cr_2Se_2O as the tunnel barrier, which is a van der Waals d -wave altermagnetic semiconductor sharing the same crystal structure as V_2Te_2O . Owing to the van der Waals nature of the interface between Cr_2Se_2O and KV_2Se_2O , the magnetic coupling across the interface is expected to be weak, thereby ensuring device stability [59]. However, due to the relatively small band gap (0.26 eV) of bulk Cr_2Se_2O , electronic states from the metallic electrodes penetrate more strongly into the barrier region, resulting in enhanced leakage in the AP state and, consequently, a substantially reduced TMR of $9.1 \times 10^4\%$.

E. Crystal facet engineering in KV_2Se_2O -based AMTJ

Due to the pronounced anisotropy of the Fermi surface in KV_2Se_2O , its transport characteristics exhibit strong directional dependence. When the current flows along the [100] direction, the symmetry constraints ($[C_2||C_{4z}]$) responsible for spin neutrality in the [001] orientation are lifted, resulting in spin-polarized transport analogous to that in conventional FM-based MTJs. To ensure structural compatibility and minimize lattice strain, the semiconductor $TiOF_2$ is chosen as the insulating barrier. The atomic and electronic structures of $TiOF_2$ are shown in Figures 5(g,h). $TiOF_2$ crystallizes in a tetragonal lattice and exhibits an excellent lattice match with KV_2Se_2O (mismatch $< 1\%$). Furthermore, $TiOF_2$ possesses a wide band gap of 2.67 eV, which remains well preserved in the $KV_2Se_2O|TiOF_2|KV_2Se_2O$ (100) heterostructure [Figure 5(i)].

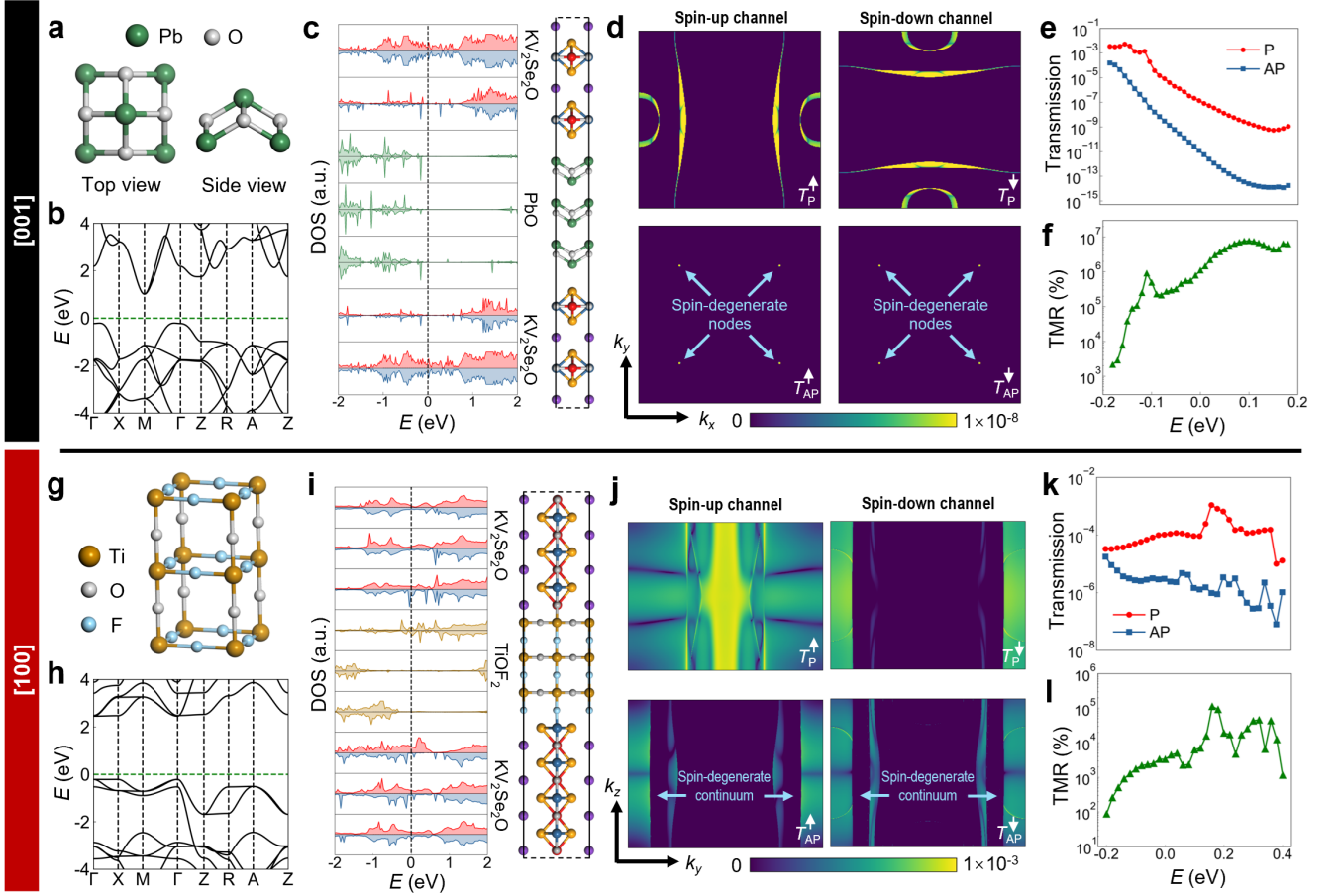


FIG. 5. Transport properties of $\text{KV}_2\text{Se}_2\text{O}$ -based AMTJs along the (a-f) [001] and (g-i) [100] directions. a) Crystal structure of the barrier material PbO . b) Electronic band structures of bulk PbO . c) Atomic structure and layer-resolved density of states (LDOS) of the $\text{KV}_2\text{Se}_2\text{O}|\text{PbO}|\text{KV}_2\text{Se}_2\text{O}$ (001) MTJ. Each panel corresponds to a single atomic layer of either $\text{KV}_2\text{Se}_2\text{O}$ or PbO , with spin-up and spin-down components shown in the upper and lower subpanels. The Fermi level is marked by a dashed line. d) $k_{\parallel}$ -resolved spin-polarized transmission coefficients at the Fermi level for the (001) AMTJ in both P and AP states. e) Total transmission spectra and f) TMR ratio as functions of energy. (g-l) Same as (a-f) but for the $\text{KV}_2\text{Se}_2\text{O}|\text{TiOF}_2|\text{KV}_2\text{Se}_2\text{O}$ MTJ along the [100] direction.

The transmission along the [100] direction [Figure 5(f)] exhibit distinct patterns compared with that along [001]. For spin-up electrons, the *quasi-2D* Fermi surface sheets extended along $k_z - k_y$ plane form continuous conduction channels throughout the BZ, resulting in a relatively high transmission. In contrast, the spin-down Fermi sheets, oriented primarily along the $k_z - k_x$ plane, have only limited projection onto the [100] transport plane. Consequently, $T_P^{\downarrow}$ is dominated by isolated spin-down Fermi pockets located near $k_y = \pm\pi/a$. In the AP configuration, the number of available transmission states is markedly reduced, with tunneling restricted to small regions near the Brillouin zone boundaries where the spin-down pockets overlap with the opposite-spin Fermi sheets. The resulting spin polarization, defined as $p = (T^{\uparrow} - T^{\downarrow})/(T^{\uparrow} + T^{\downarrow})$, reaches 73.2% when the Néel vectors of the $\text{KV}_2\text{Se}_2\text{O}$ electrodes are aligned.

By integrating over $k_{\parallel}$, we obtain the total transmission spectra and corresponding TMR ratio as functions of

energy [Figures 5(k,l)]. Because both the P and AP configurations host a larger number of available conduction channels, the overall transmission along the [100] direction is substantially enhanced. At the Fermi energy, the TMR reaches $3.3 \times 10^3\%$, which is much smaller than that of the [001] device due to the spin-degenerate continua that yield sizable AP-state tunneling. Here, [100] configurations have the advantage of overall larger transmission value in the P states, hence better signal-to-noise ratio during device readout [72], but this comes at the cost of a reduced TMR relative to the [001] counterpart. The pronounced contrast between the [001] and [100] configurations thus underscores the crystal facet engineering in tailoring the performance metrics of AMTJs.

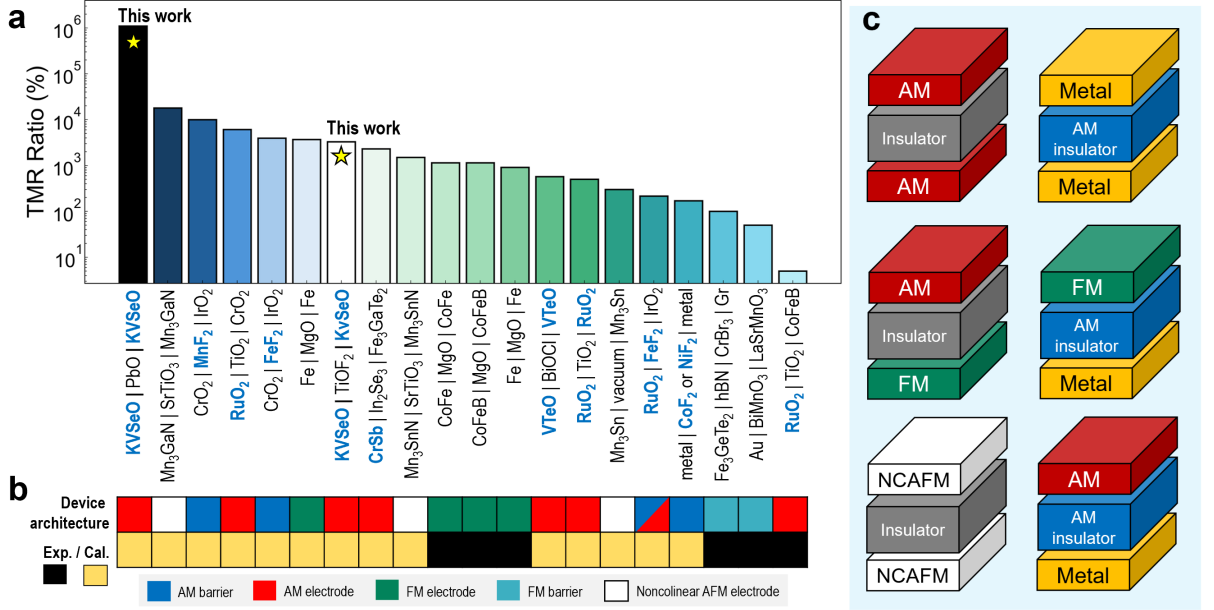


FIG. 6. Benchmark of MTJ performances. a) Comparison of TMR ratios obtained using KV₂Se₂O electrodes in this work (highlighted with stars) and selected results from previous experimental and theoretical studies [21, 25, 35, 36, 56, 57, 60–71]. To ensure a consistent comparison, all experimental values listed here are measured at low temperatures. Altermagnetic materials are marked with blue. For simplicity, KV₂Se₂O and V₂Te₂O are abbreviated as KVSeO and VTeO, respectively and “Gr” denotes graphene. b) Device architectures and methodology (experimental or computational) corresponding to each bar in panel (a). The top row codes the type of functional layer – AM barrier, AM electrode, FM electrode, FM barrier or noncollinear AFM (NCAFM) electrode, while the bottom row distinguishes between experiment (Exp.) and calculation (Cal.). c) Schematic illustrations of representative MTJ architectures with AM or NCAFM.

F. Performance Benchmarking

We provide a performance benchmark of KV₂Se₂O-based AMTJs in Figure 6. Their TMR ratios are compared with representative experimental and theoretical results from recent MTJ studies [Figs. 6a,b; see Section F of the Supporting Information for details] under different device configurations [see Figure 6(c) for representative MTJ setups]. We note that MTJs based on multilayer CrI₃ tunnel barriers, although reported to exhibit giant TMR, rely on a magnetic-field-induced phase transition of the barrier [73–75] and its device operation is restricted to cryogenic temperatures due to the low critical temperature of CrI₃ [76]. Owing to this fundamentally different mechanism and limited operating temperature window, such devices are not included in our benchmark.

Under typical MTJ operating conditions with low applied bias voltages (on the order of 10^{−1} V), the transport characteristics are primarily governed by the transmission characteristics at the Fermi level [77, 78]. Therefore, for quantitative benchmarking and direct comparison, we focus on the TMR values evaluated at E_F for simulation results. Conventional FM MTJs, such as Fe|MgO|Fe, typically yield theoretical TMR values around 3700% [63], whereas the highest experimentally reported TMR values have reached 417% at room temperature (RT) and 914% at 3 K [69]. For

CoFe|MgO|CoFe or CoFeB|MgO|CoFeB devices, higher TMR ratios have been achieved recently, exceeding 600% at RT and 1100% at low temperatures [70, 71]. In comparison, the simulated TMR ratios of RuO₂-based AMTJs remain on the order of 10³% [21, 62, 66].

Notably, our proposed KV₂Se₂O|PbO|KV₂Se₂O [001] device achieves a record-high TMR of $1.1 \times 10^6\%$, surpassing all previously reported AMTJs here by more than two orders of magnitude. This exceptional performance originates from the combination of: (i) the altermagnetic flatband Fermi surface geometry of KV₂Se₂O which yields minimal nodal-like conduction channels in the AP state; combined with (ii) the symmetry-matched PbO insulating barrier. When transport occurs along the [100] direction, the TMR is substantially reduced due to the presence of extended spin-matched conduction channels in the AP configuration – though the resulting TMR remains comparable to many other MTJ designs. Overall, our computational results establish the theoretical performance ceiling of KV₂Se₂O-based AMTJs, providing both a quantitative benchmark for material selection and a conceptual framework for the Fermi-surface-engineered design of next-generation spintronic devices.

III. CONCLUSION

In summary, we have systematically investigated the intrinsic transport characteristics in AMTJs using three experimentally realized bulk altermagnets – V_2Te_2O , RbV_2Te_2O , and KV_2Se_2O . We demonstrated that the degree of overlap between opposite-spin conduction channels serves as the key factor governing the TMR performance. Among these materials, KV_2Se_2O exhibits nearly nodal-like spin-degenerate conduction channels in the AP state, which arises from its altermagnetic flatband Fermi surface geometry. As a result, an intrinsic TMR of $4.3 \times 10^3\%$ was obtained, which was further enhanced to $1.1 \times 10^6\%$ by employing a symmetry-matched PbO insulating barrier in an AMTJ device setup. The pronounced contrast between the [001] and [100] transport directions underscores the essential role of Fermi surface geometry engineering via crystallographic orientation, which enables the overall magnitude of the transmission and the TMR to be tuned and balanced depending on the application needs.

Beyond identifying KV_2Se_2O as a compelling altermagnetic electrode material, our results revealed the pivotal role of altermagnetic flatbands in achieving ultrahigh-TMR AMTJs with minimal spin-degenerate “arc-” or “node-” like conduction channels. More broadly, this work established a concrete design principle for altermagnetic-spintronic devices: high-performance AMTJs are enabled when quasi-layered altermagnets hosting flatband-driven, symmetry-protected Fermi surface geometries that minimize opposite-spin overlap. This flatband-centric paradigm provides a powerful materials design and screening guideline and paves a pathway toward next-generation altermagnetic-spintronics uniquely empowered by quasi-layered altermagnets.

IV. EXPERIMENTAL SECTION

A. Density Functional Theory Simulations

The geometry optimization and electronic structure calculations for V_2Te_2O , RbV_2Te_2O and KV_2Se_2O are carried out using the Vienna Ab initio Simulation Package (VASP) [79, 80]. The projector augmented wave (PAW) method is employed, with the exchange-correlation effects treated within the generalized gradient approximation (GGA) using the Perdew–Burke–Ernzerhof (PBE) functional [81]. We set the plane-wave cutoff energy to 600 eV and the atomic positions are relaxed until the residual forces on each atom are less than 10^{-2} eV/Å. The Brillouin zone is sampled using dense Monkhorst–Pack k -point grids of $25 \times 25 \times 13$ for bulk material unit-cell calculations. For AMTJ device optimizations, $23 \times 23 \times 1$ and $23 \times 13 \times 1$ meshes are employed for transport along the [001] and [100] directions, respectively. The on-site Coulomb interaction is treated within the GGA+ U scheme [82, 83],

with $U_{\text{eff}} = 3.7$ eV on V 3d orbitals for V_2Te_2O [49] and 1.0 eV for RbV_2Te_2O [32]. We do not apply a Hubbard U correction when calculating KV_2Se_2O , as our tests [see Section A of the Supporting Information for details] show that this setting yields band structures most consistent with the experimental ARPES results [29]. To minimize lattice mismatch and the associated structural distortion, we employ PbO and $TiOF_2$ as the tunnel barriers for the KV_2Se_2O AMTJ along the [001] and [100] directions, respectively and we apply a Hubbard U correction of 4.2 eV on Ti 3d orbitals [84].

B. Quantum Atomistic Device Simulations

The spin-dependent transport properties are evaluated using Density Functional Theory (DFT) combined with the nonequilibrium Green’s function (NEGF) formalism, implemented in the QuantumATK package [85, 86]. Our device is modeled as a two-probe system, consisting of a central region and two electrodes. The spin-resolved transmission coefficient $T_{k_{\parallel},s}(E)$ can be written as

$$T_{k_{\parallel},s}(E) = \text{Tr}[G_{k_{\parallel},s}(E)\Gamma_{k_{\parallel},s}^l(E)G_{k_{\parallel},s}(E)^{\dagger}\Gamma_{k_{\parallel},s}^r(E)]$$

where $k_{\parallel}$ denotes the reciprocal lattice vector parallel to the electrode interface (orthogonal to the transport direction) in the irreducible Brillouin zone, and s labels the spin. Here, $G_{k_{\parallel},s}(E)$ is the retarded Green’s function for spin s at $k_{\parallel}$, while $\Gamma_{k_{\parallel},s}^{l(r)}(E)$ denotes the broadening function of the left (right) electrode, which equals the imaginary part of the corresponding self-energy term [87]. The AMTJ simulations employ a density mesh cutoff of 180 Hartree and an electronic temperature of 300 K. For the self-consistent calculations, the Brillouin zone is sampled using k -point meshes of $23 \times 23 \times 1$ and $23 \times 13 \times 1$ for devices oriented along the [001] and [100] directions, respectively. For the transmission calculations, much denser k -point meshes of 301×301 and 201×101 are used to reliably resolve the minimal spin-channel overlap. To accurately describe the van der Waals interaction in V_2Te_2O multilayers, Grimme’s DFT-D3 correction is applied [88].

C. Statistical Analysis

The results presented in this work are obtained from first-principles electronic structure and quantum transport calculations. No statistical methods or hypothesis testing were employed, as the calculations do not involve stochastic sampling or experimental variability. All data are presented as direct computational outputs without preprocessing, normalization, or outlier treatment. Sample size, measures of variance, and statistical significance testing are therefore not applicable. All simulations and analyses were carried out using VASP, QuantumATK and post-processed with Python-based analysis tools.

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